

Linear time functional dynamic programming for un-regularized isotonic regression

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1 Introduction

Isotonic regression is a statistical model which fits a non-decreasing function to a sequence of n data. Previous algorithms for computing this model include the Pool Adjacent Violators Algorithm (PAVA), which can be implemented in linear time, $O(n)$ [Busing, 2022]. We propose a new dynamic programming algorithm, and demonstrate the similarities and differences with PAVA.

2 Problem

2.1 Data

We assume there are n data, $y_1, \dots, y_n \in \mathbb{R}$. In some contexts, there are also positive weights $w_1, \dots, w_n \in \mathbb{R}_+$. When no weights are given, we take $w_i = 1$ for all i .

2.2 Isotonic regression problem

The weighted isotonic regression problem is

$$\hat{C}_n = \min_{x_1, \dots, x_n} \sum_{i=1}^n w_i (x_i - y_i)^2 \text{ subject to } \forall i \in \{2, \dots, n\}, x_{i-1} \leq x_i. \quad (1)$$

The $\hat{C}_n \in \mathbb{R}$ is the optimal cost, including all n data points.

3 Algorithms

3.1 Proposed dynamic programming algorithm

We propose a new algorithm based on previous work from the literature on optimal change-point detection [Hocking et al., 2020]. The idea is to use dynamic programming to recursively

compute cost functions,

$$C_n(x_n) = \min_{x_1, \dots, x_{n-1}} \sum_{i=1}^n w_i (x_i - y_i)^2 \text{ subject to } \forall i \in \{2, \dots, n\}, x_{i-1} \leq x_i. \quad (2)$$

$$= w_n (x_n - y_n)^2 + \min_{x_{n-1}} C_{n-1}(x_{n-1}) \text{ subject to } x_{n-1} \leq x_n. \quad (3)$$

Algorithm 1 summarizes the solution method for the simple case without weights ($w_i = 1$ for all i). Whereas previous functional dynamic programming algorithms have used either a linked list or binary tree data structure, the proposed algorithm uses arrays, which is a much simpler design that is only possible because of the problem structure. $C_i(x)$ is a convex function, defined in terms of pieces, each piece corresponding to a cluster of data with the same mean, and the min is always on the last piece (TODO examples and proof).

Algorithm 1 Functional dynamic programming for isotonic regression

Require: data $y_1, \dots, y_n \in \mathbb{R}$.

Allocate cluster sizes $N \in \mathbb{Z}^n$ and totals/means $T, M \in \mathbb{R}^n$.

Initialize $K \leftarrow 1$, $N_1 \leftarrow 1$, $T_1 \leftarrow y_1$, $M_1 = y_1$.

for $i = 2$ to n **do**

samples $\leftarrow 1$, **total** $\leftarrow y_i$.

while $K > 0$ and **total**/**samples** $< M_K$ **do**

samples $+= N_K$, **total** $+= T_K$, $K --$.

end while

$K ++$, $N_K \leftarrow \text{samples}$, $T_K \leftarrow \text{total}$, $M_K \leftarrow \text{total}/\text{samples}$.

end for

return K, N, M

Algorithm 1 outputs the optimal means in a run-length encoding, for a number of clusters $K \in \{1, \dots, n\}$. Algorithm 2 can be used to convert into the solution x of problem (1), the n -vector with an optimal mean value for each input data point.

Algorithm 2 Decoding optimal isotonic regression parameters

Require: output of Algorithm 1: number of clusters $K \in \mathbb{Z}$, sizes $N \in \mathbb{Z}^n$, means $M \in \mathbb{R}^n$.

Allocate optimal means to output, $x \in \mathbb{R}^n$.

Initialize $i \leftarrow n$.

for $k = K$ to 1 **do**

for $j = 1$ to N_k **do**

$x_i \leftarrow M_k$, $i --$.

end for

end for

return x

3.2 Comparison with PAVA

TODO

3.3 Extensions for bounds and weights

TODO

References

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